Lab 7: The Link Layer in Practice

Ethernet Frames, MAC Addresses, and ARP

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Welcome to Lab 7: Bridging Theory and Reality

- **Where we are:** In Chapter 5, we looked at the Network Layer (IP addresses and global routing). Today, we move down to Chapter 6: The Link Layer.
- **The Goal of this Lab:** An IP address gets your data to the right network. But to physically move data across a copper wire or Wi-Fi signal to a specific machine in this room, we need the Link Layer.
- **What we will do:** I will walk you through the core concepts, and then you will use Google Colab's Linux terminal and Wireshark on your own machines to see these physical addresses in real-time.

Concept 1: What is a MAC Address? (Section 6.4.1)

To communicate on a local network, IP addresses are not enough. We need physical hardware addresses.

- **Acronym Definition:** MAC stands for **Media Access Control**.
- **What is it?** It is a 48-bit physical address. You will usually see it written as 6 pairs of hexadecimal letters and numbers (for example: `1A:2B:3C:4D:5E:6F`).
- **It is Permanent:** An IP address changes depending on what coffee shop or campus you connect to. A MAC address never changes. It is permanently burned into your computer's **NIC (Network Interface Controller)** at the factory.
- **Analogy:** Your MAC address is your Social Security Number. Your IP address is your current postal mailing address.

Lab Part 1: Finding Your MAC Address

In the first part of today's lab, you will find the MAC address of a cloud computer.

- **The Tool:** You will open a Google Colab notebook and use the `!ip link show` command in the terminal.
- **What to look for:** You will see an interface called `eth0` (Ethernet zero). Look for the phrase `link/ether`. The 48-bit number next to it is the physical MAC address of that cloud server.
- **The Loopback Exception:** You will also see an interface called `lo` (loopback). This is a virtual interface computers use to talk to themselves (IP 127.0.0.1). Because it never touches a physical wire, it does not have a standard MAC address!

Concept 2: The Translation Problem

The Problem We Must Solve

Imagine your computer wants to send a webpage request to the local campus router at IP `192.168.1.1`. Your computer's network card only understands MAC addresses, not IP addresses. How does your computer figure out the router's physical MAC address?

- **The Solution:** We use a protocol called **ARP**.
- **Acronym Definition:** ARP stands for **Address Resolution Protocol**.
- **What it does:** It simply translates a known IP address into an unknown MAC address so the computer can build the Ethernet frame and put it on the wire.

How ARP Works: The "Shouting" Analogy (Section 6.4.1)

ARP works exactly like trying to find someone in a crowded room.

1. **The ARP Request (The Shout):** Your computer sends a broadcast message to every single machine on the local network. It essentially shouts: *"Hey everyone! Who has the IP address 192.168.1.1? Please tell me your MAC address!"*
2. **The ARP Reply (The Whisper):** Every machine hears the shout, but only the machine with that specific IP address replies. It sends a direct, private message back saying: *"That's me! My MAC address is AA:BB:CC:DD:EE:FF."*
3. **The ARP Cache (The Memory):** Your computer saves this answer in its memory (called the ARP Table) so it doesn't have to shout again the next time it wants to talk to the router.

Lab Part 2: Inspecting the ARP Table

In the second part of the lab, you will look at your computer's memory.

- **The Tool:** In Google Colab, you will run the `!arp -a` command. This prints out the cached IP-to-MAC translations.
- **The Off-Network Rule:** You will then ping Google's DNS at `8.8.8.8` and check the ARP table again.
- **Crucial Concept:** You will *not* see Google's MAC address in your table. Why? Because ARP only works on your **local physical subnet**. To reach Google, your computer actually ARPs for your **Default Gateway Router's** MAC address, and hands the packet to the router to send across the Internet.

Lab Part 3: Capturing ARP in Wireshark

In the final part of the lab, we will prove this process happens by catching the "shout" and the "whisper" out of the air using Wireshark.

- **Step 1:** You will clear your computer's ARP cache. This gives your computer amnesia, forcing it to "shout" the next time it needs to send a packet.
- **Step 2:** You will start Wireshark and ping your home or campus router.
- **Step 3:** You will filter Wireshark for `arp` and look at the Ethernet II header.

What to Look For in Wireshark (Section 6.4.2)

When you expand the Ethernet header in Wireshark, pay close attention to the Destination MAC address.

- **The ARP Request (Broadcast):** To ensure every network card in the room listens to the "shout," the sender uses a special broadcast MAC address: `FF:FF:FF:FF:FF:FF`. If you see all F's, it is a broadcast.
- **The ARP Reply (Unicast):** Look at the reply packet. The destination MAC will no longer be all F's. It will be the specific, unique MAC address of your computer, because the router is answering you directly.

Final Concept: Why do we need this for Switches?

Why do we go through all this trouble to find MAC addresses? Because of local network switches.

- **Layer 2 Switches:** The network switches in the ceilings of this building do not understand IP addresses. They only understand MAC addresses.
- **Self-Learning:** When your computer sends an Ethernet frame, the switch looks at your **Source MAC address** and remembers which physical cable you are plugged into.
- **Forwarding:** It then looks at the **Destination MAC address** (which you found using ARP) and forwards the frame exactly down the correct cable to reach the router.

Let's begin the lab! Open your handouts and start Part 1.